

Task 1

derive (BCE) loss

Binary Cross-Entropy Loss

$$J = -\frac{1}{n} \sum_{i=1}^n [y^{(i)} \log(\hat{y}^{(i)}) + (1-y^{(i)}) \log(1-\hat{y}^{(i)})]$$

Logic Regression:-

$$\hat{y} = \sigma(Z) \quad \text{where} \quad Z^{(i)} = \omega^T x + b \quad \sum_{j=1}^n \omega_j x_j^{(i)} + b$$
$$Z = \omega^T x + b$$

$$\frac{\partial J}{\partial \omega_i} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial Z} \cdot \frac{\partial Z}{\partial \omega_i}$$

$$\frac{\partial J}{\partial \hat{y}} = [-y \log(\hat{y}) - (1-y) \log(1-\hat{y})]$$
$$-y \times \frac{1}{\hat{y}} - (1-y) \times \frac{-1}{(1-\hat{y})}$$

$$\frac{\partial \hat{y}}{\partial Z} = y(1-\hat{y})$$

~~for~~ $Z = \omega_1 x_1 + \dots + \omega_j x_j + \dots + b$

$$\frac{\partial Z}{\partial \omega_j} = x_j$$

$$\frac{\partial J}{\partial \omega_i} = \left(-\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}} \right) \hat{y}(1-\hat{y}) x_j$$

$$= \hat{y}(1-\hat{y}) x_j \left(-\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}} \right)$$

$$= -y(1-\hat{y}) x_j + \hat{y}(1-y) x_j$$

$$= (-y + y\hat{y} + \hat{y} - y\hat{y}) x_j = (\hat{y} - y) x_j$$

$$\Rightarrow \frac{\partial J}{\partial \omega_i} = \frac{1}{n} \sum_{i=1}^n (y^{(i)} - \hat{y}^{(i)}) x_j$$